

UNIT- IV

VECTOR (14 Marks)

- Vector is denoted by $\vec{x} = (x_1, x_2, x_3) = x_1\hat{i} + x_2\hat{j} + x_3\hat{k}$.
- Modulus of vector is defined as
- $$|\vec{x}| = \sqrt{x_1^2 + x_2^2 + x_3^2}$$
- If $\vec{x} = (x_1, x_2, x_3)$ and $\vec{y} = (y_1, y_2, y_3)$ then Addition & Subtraction of these two vectors is
- $$\vec{x} + \vec{y} = (x_1 + y_1, x_2 + y_2, x_3 + y_3) \text{ and}$$
- $$\vec{x} - \vec{y} = (x_1 - y_1, x_2 - y_2, x_3 - y_3).$$
- Unit vector If $|\vec{x}| = 1$, then vector $\vec{x}$ is called Unit vector.
- Direction cosines of vector :-
- $$l = \frac{x}{\sqrt{x^2 + y^2 + z^2}}, \quad m = \frac{y}{\sqrt{x^2 + y^2 + z^2}}, \quad n = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$
- Dot product of two vectors
- $$\vec{x} \cdot \vec{y} = (x_1, x_2, x_3) \cdot (y_1, y_2, y_3)$$
- $$= x_1 y_1 + x_2 y_2 + x_3 y_3.$$
- $\vec{x} \cdot \vec{y} = \vec{y} \cdot \vec{x}$
- Cross product of two vectors
- $$\vec{x} \times \vec{y} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$
- $$\vec{i} = (1, 0, 0)$$
- $$\vec{j} = (0, 1, 0)$$
- $$\vec{k} = (0, 0, 1)$$
- $\vec{x} \times \vec{y} = -\vec{y} \times \vec{x}$

* Angle between two vectors

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

and

$$|\vec{x} \times \vec{y}| = |\vec{x}| |\vec{y}| \sin \theta$$

* Work-done by Force

$$W = \vec{F} \cdot \vec{d}, \quad \text{where } \vec{F} = \vec{F}_1 + \vec{F}_2 = \text{Total Force}$$

$$\vec{d} = d_2 - d_1 = \text{displacement} \\ \text{To point - From point}$$

* perpendicular unit vectors

→ To find perpendicular unit vectors.

$$\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

→ If $\vec{a}$ & $\vec{b}$ are perpendicular to each other then

$$\vec{a} \cdot \vec{b} = 0$$

→ If $\vec{a}$

* Question for 1 Marks :-

1. If $\vec{a} = 2\hat{i} + 3\hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - 3\hat{j} + 2\hat{k}$ then $\vec{a} + \vec{b} = \underline{\hspace{2cm}}$

$$\text{Here } \vec{a} = (2, 3, 1), \quad \vec{b} = (2, -3, 2)$$

$$\vec{a} + \vec{b} = (2, 3, 1) + (2, -3, 2)$$

$$= (4, 0, 3)$$

$$\therefore \boxed{\vec{a} + \vec{b} = 4\hat{i} + 3\hat{k}}$$

2> If $\vec{a} = 2\vec{i} + 3\vec{j}$, $\vec{b} = 3\vec{i} - \vec{j} - 2\vec{k}$ then $\vec{a} - \vec{b}$ is.

Here $\vec{a} = (2, 3, 0)$

$$\vec{b} = (3, -1, -2)$$

$$\vec{a} - \vec{b} = (2, 3, 0) - (3, -1, -2)$$

$$= (2-3, 3+1, 0+2)$$

$$= (-1, 4, 2)$$

$$\therefore \vec{a} - \vec{b} = -\vec{i} + 4\vec{j} + 2\vec{k}.$$

3> If $\vec{a} = -\vec{i} + 3\vec{j}$ then $|\vec{a}| = \underline{\hspace{2cm}}$

Here $\vec{a} = -\vec{i} + 3\vec{j}$

$$\vec{a} = (-1, 3)$$

$$|\vec{a}| = \sqrt{(-1)^2 + 3^2} = \sqrt{1+9} = \sqrt{10}$$

$$\therefore \boxed{|\vec{a}| = \sqrt{10}}$$

4> If $\vec{u} = \frac{1}{\sqrt{5}}\vec{i} + \frac{2}{\sqrt{5}}\vec{j}$ then $|\vec{u}| = \underline{\hspace{2cm}}$

Here $\vec{u} = \frac{1}{\sqrt{5}}\vec{i} + \frac{2}{\sqrt{5}}\vec{j}$

$$\vec{u} = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$$

$$\therefore |\vec{u}| = \sqrt{\left(\frac{1}{\sqrt{5}}\right)^2 + \left(\frac{2}{\sqrt{5}}\right)^2} = \sqrt{\frac{1}{5} + \frac{4}{5}} = \sqrt{\frac{1+4}{5}} = \sqrt{\frac{5}{5}} = \sqrt{1} = 1.$$

$$\therefore \boxed{|\vec{u}| = 1}$$

5> If $\vec{a} = 3\vec{i} - 4\vec{j} - 5\sqrt{3}\vec{k}$ then $|\vec{a}| = \underline{\hspace{2cm}}$

Here $\vec{a} = (3, -4, -5\sqrt{3})$

$$\text{so, } |\vec{a}| = \sqrt{(3)^2 + (-4)^2 + (-5\sqrt{3})^2}$$

$$= \sqrt{9 + 16 + 75}$$

$$= \sqrt{100}$$

$$\therefore \boxed{|\vec{a}| = 10}$$

6) If $\vec{a} = -\hat{i} + 3\hat{j}$ and $\vec{b} = 2\hat{i} + 3\hat{j}$, then $|\vec{a}| + |\vec{b}| = \sqrt{10} + \sqrt{13}$

Here $\vec{a} = (-1, 3)$ & $\vec{b} = (2, 3)$

$$|\vec{a}| = \sqrt{(-1)^2 + 3^2} = \sqrt{1+9} = \sqrt{10}$$

$$|\vec{b}| = \sqrt{(2)^2 + (3)^2} = \sqrt{4+9} = \sqrt{13}$$

$$\therefore |\vec{a}| + |\vec{b}| = \sqrt{10} + \sqrt{13}$$

7) If $\vec{a} = 2\hat{i} - 3\hat{j}$, $\vec{b} = 3\hat{j} - 4\hat{k}$ and $\vec{c} = 4\hat{k} - 2\hat{i}$ then

$$\vec{a} + \vec{b} + \vec{c} = \underline{0}$$

Here $a = (2, -3, 0)$, $b = (0, 3, -4)$, $c = (-2, 0, 4)$

$$\therefore \vec{a} + \vec{b} + \vec{c} = (2, -3, 0) + (0, 3, -4) + (-2, 0, 4)$$

$$= (2+0-2, -3+3+0, 0-4+4)$$

$$= (0, 0, 0)$$

$$= 0 \text{ (zero vector)}$$

8) If $\vec{a} = 2\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} - 3\hat{k}$ then $\vec{a} \cdot \vec{b} = \underline{2}$.

Here $\vec{a} = (2, 1, 0)$, $\vec{b} = (1, 0, -3)$

$$\text{Then } \vec{a} \cdot \vec{b} = (2, 1, 0) \cdot (1, 0, -3)$$

$$= 2 + 0 + 0$$

$$= 2.$$

$$\therefore \boxed{\vec{a} \cdot \vec{b} = 2}$$

9) If $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 3\hat{k}$ then $\vec{a} \cdot \vec{b} = \underline{4}$

$\vec{a} = (2, 1, 1)$, $\vec{b} = (1, -1, 3)$

$$\text{Then } \vec{a} \cdot \vec{b} = (2, 1, 1) \cdot (1, -1, 3)$$

$$= 2 - 1 + 3$$

$$= 4$$

$$\therefore \boxed{\vec{a} \cdot \vec{b} = 4}$$

10> If $\vec{a} = 2\vec{i} - 2\vec{j} + \vec{k}$ and $\vec{b} = \vec{i} + 3\vec{j} + 3\vec{k}$ then $\vec{a} \cdot \vec{b} = \underline{-1}$

Here $\vec{a} = (2, -2, 1)$

$\vec{b} = (1, 3, 3)$

$$\text{Then } \vec{a} \cdot \vec{b} = (2, -2, 1) \cdot (1, 3, 3) \\ = 2 - 6 + 3$$

$$\therefore \boxed{\vec{a} \cdot \vec{b} = -1}$$

11> If $\vec{a} = 2\vec{i} + 3\vec{j} - \vec{k}$ and $\vec{b} = 4\vec{i} + 6\vec{j} - 2\vec{k}$ then $\vec{a} \times \vec{b} = \underline{0}$

→ Here $\vec{a} = (2, 3, -1)$ & $\vec{b} = (4, 6, -2)$

Then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & -1 \\ 4 & 6 & -2 \end{vmatrix}$$

$$= \vec{i}[-6 + 6] - \vec{j}[-4 + 4] + \vec{k}[12 - 12]$$

$$= 0\vec{i} - 0\vec{j} + 0\vec{k}$$

$$\boxed{\vec{a} \times \vec{b} = 0} \text{ (zero vector)}$$

12> If $\vec{a} = \vec{i} + 3\vec{j} - \vec{k}$, $\vec{b} = 4\vec{i} + \vec{j} - 2\vec{k}$ then $\vec{a} \times \vec{b} = \underline{-(5\vec{i} + 2\vec{j} + 11\vec{k})}$.

Here $\vec{a} = (1, 3, -1)$

$\vec{b} = (4, 1, -2)$

Then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & -1 \\ 4 & 1 & -2 \end{vmatrix}$$

$$= \vec{i}(-6 + 1) - \vec{j}(-2 + 4) + \vec{k}(1 - 12)$$

$$= \vec{i}(-5) - \vec{j}(2) + \vec{k}(-11)$$

$$= (-5, -2, -11)$$

$$= -(5\vec{i} + 2\vec{j} + 11\vec{k})$$

13> If $\vec{a} = \vec{i} + \vec{j}$ and $\vec{b} = \vec{j} - \vec{i}$ then Angle $(\vec{a}, \vec{b}) = \underline{\pi/2}$

Here $\vec{a} = (1, 1)$ & $\vec{b} = (-1, 1)$

$$|\vec{a}| = \sqrt{1+1} = \sqrt{2}, \quad |\vec{b}| = \sqrt{1+1} = \sqrt{2}$$

$$\begin{aligned}\text{Then } \bar{a} \cdot \bar{b} &= (1, 1) \cdot (-1, 1) \\ &= -1 + 1 \\ &= 0\end{aligned}$$

$$\therefore \bar{a} \cdot \bar{b} = 0.$$

$$\therefore \cos \theta = \frac{\bar{a} \cdot \bar{b}}{|\bar{a}| |\bar{b}|} = \frac{0}{\sqrt{2} \sqrt{2}} = 0$$

$$\therefore \theta = \cos^{-1} 0 \Rightarrow \boxed{\theta = \pi/2}$$

14) If $\bar{a} = 2\hat{i} - 3\hat{j}$, $\bar{b} = \hat{i} - 3\hat{j}$ and $\bar{c} = 3\hat{i} + \hat{j}$ then

$$2\bar{a} - (\bar{b} + \bar{c}) = \underline{0 - 4\hat{j}}.$$

Here $a = (2, -3)$, $b = (1, -3)$, $c = (3, 1)$

Now, $2\bar{a} - (\bar{b} + \bar{c})$

$$= 2(2, -3) - [(1, -3) + (3, 1)]$$

$$= (4, -6) - (4, -2)$$

$$= (4 - 4, -6 + 2)$$

$$= (0, -4)$$

$$2\bar{a} - (\bar{b} + \bar{c}) = 0 - 4\hat{j}$$

15) If $\bar{a} = \hat{i} + 3\hat{j}$ and $\bar{b} = 5\hat{i} - \hat{j}$ then $|\bar{a} + 3\bar{b}| = \underline{16}$

Here $a = (1, 3)$, $b = (5, -1)$

$$\bar{a} + 3\bar{b} = (1, 3) + 3(5, -1)$$

$$= (1, 3) + (15, -3)$$

$$= (16, 0)$$

$$\therefore |\bar{a} + 3\bar{b}| = \sqrt{(16)^2 + 0} = \sqrt{256}$$

$$\therefore |\bar{a} + 3\bar{b}| = 16$$

16) If $\vec{x} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ then $|\vec{x}| = \underline{\hspace{2cm}}$

Here $\vec{x} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

Then $|\vec{x}| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2}$
 $= \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{\frac{2}{2}} = \sqrt{1}$

$\therefore \boxed{|\vec{x}| = 1}$

17) $\vec{a} \times \vec{a} = \underline{0}$ (zero vector)

Let $\vec{a} = (1, 1, 1)$

Then

$$\vec{a} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= \hat{i}(1-1) - \hat{j}(1-1) + \hat{k}(1-1)$$

$$= 0\hat{i} - 0\hat{j} + 0\hat{k}$$

$$= (0, 0, 0)$$

$\therefore \boxed{\vec{a} \times \vec{a} = 0}$

18) $\vec{a} \cdot \vec{a} = \underline{|\vec{a}|^2}$

19) $\vec{a} \cdot (\vec{a} \times \vec{b}) = \underline{0}$

Suppose $\vec{a} = (1, 2, 3)$ & $\vec{b} = (3, 1, 2)$.

Then $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{vmatrix}$

$$= \hat{i}(4-3) - \hat{j}(2-9) + \hat{k}(1-6)$$

$$\vec{a} \times \vec{b} = \hat{i} + 7\hat{j} - 5\hat{k}$$

$$\vec{a} \times \vec{b} = (1, 7, -5)$$

$$\vec{a} \cdot (\vec{a} \times \vec{b}) = (1, 2, 3) \cdot (1, 7, -5)$$

$$= 1 + 14 - 15 = 0$$

$$\therefore \boxed{\vec{a} \cdot (\vec{a} \times \vec{b}) = 0}$$

$$20) (\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{a}) = \underline{|\vec{a} \times \vec{b}|^2}$$

$$\rightarrow \vec{b} \times \vec{a} = -(\vec{a} \times \vec{b}).$$

$$(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{a}) = -(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b})$$

$$= -|\vec{a} \times \vec{b}|^2$$

* Question for 3 Marks

1) If $\vec{a} = 2\hat{i} + \hat{j} - 3\hat{k}$, $\vec{b} = 4\hat{i} + 5\hat{j} + 4\hat{k}$ and $\vec{c} = 3\hat{i} - 2\hat{j} + \hat{k}$
then find $3\vec{a} + 2\vec{b} - 3\vec{c}$.

$\rightarrow$ Here $\vec{a} = (2, 1, -3)$, $\vec{b} = (4, 5, 4)$, $\vec{c} = (3, -2, 1)$

Then $3\vec{a} + 2\vec{b} - 3\vec{c}$

$$= 3(2, 1, -3) + 2(4, 5, 4) - 3(3, -2, 1)$$

$$= (6, 3, -9) + (8, 10, 8) - (9, -6, 3)$$

$$= (6+8-9, 3+10+6, -9+8-3)$$

$$= (5, 19, -4)$$

$$3\vec{a} + 2\vec{b} - 3\vec{c} = 5\hat{i} + 19\hat{j} - 4\hat{k}.$$

2) If $\vec{a} = \hat{i} - 2\hat{j} + 4\hat{k}$, $\vec{b} = 3\hat{i} + \hat{j} - 4\hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} - 4\hat{k}$
then find $5\vec{a} + 3\vec{b} + 2\vec{c}$.

$\rightarrow$ Here $\vec{a} = (1, -2, 4)$, $\vec{b} = (3, 1, -4)$, $\vec{c} = (1, 2, -4)$

Then $5\vec{a} + 3\vec{b} + 2\vec{c}$

$$= 5(1, -2, 4) + 3(3, 1, -4) + 2(1, 2, -4)$$

$$= (5, -10, 20) + (9, 3, -12) + (2, 4, -8)$$

$$= (5-9+2, -10+3+4, 20-12-8)$$

$$= (-2, -3, 0)$$

$$\rightarrow |5\bar{a} + 3\bar{b} + 2\bar{c}| = \sqrt{(-2)^2 + (-3)^2 + 0^2}$$

$$= \sqrt{4+9}$$

$$\therefore |5\bar{a} + 3\bar{b} + 2\bar{c}| = \sqrt{13}$$

3) If $\bar{a} = \hat{j} + \hat{k} - \hat{l}$ and $\bar{b} = 2\hat{i} + \hat{j} - 3\hat{k}$ then find $|2\bar{a} + 3\bar{b}|$.

$$\rightarrow \text{Here } \bar{a} = (-1, 1, 1)$$

$$\bar{b} = (2, 1, -3)$$

$$\text{Then } 2\bar{a} + 3\bar{b} = 2(-1, 1, 1) + 3(2, 1, -3)$$

$$= (-2, 2, 2) + (6, 3, -9)$$

$$= (-2+6, 2+3, 2-9)$$

$$= (4, 5, -7)$$

$$\therefore |2\bar{a} + 3\bar{b}| = \sqrt{4^2 + 5^2 + (-7)^2}$$

$$= \sqrt{16 + 25 + 49}$$

$$= \sqrt{90}$$

$$\therefore |2\bar{a} + 3\bar{b}| = 3\sqrt{10}$$

4) If $\bar{a} = 3\hat{i} - \hat{j} - 4\hat{k}$, $\bar{b} = -2\hat{i} + 4\hat{j} - 3\hat{k}$ and $\bar{c} = -\hat{i} + 2\hat{j} - 5\hat{k}$ then find direction cosines of $\bar{a} + 2\bar{b} - \bar{c}$.

$$\rightarrow \text{Here } \bar{a} = (3, -1, -4), \bar{b} = (-2, 4, -3), \bar{c} = (-1, 2, -5)$$

$$\text{Then } \bar{a} + 2\bar{b} - \bar{c}$$

$$= (3, -1, -4) + 2(-2, 4, -3) - (-1, 2, -5)$$

$$= (3, -1, -4) + (-4, 8, -6) - (-1, 2, -5)$$

$$= (3-4+1, -1+8-2, -4-6+5)$$

$$= (0, 5, -5)$$

$$\begin{aligned}\therefore |\vec{a} + 2\vec{b} - \vec{c}| &= \sqrt{0^2 + 5^2 + (-5)^2} \\ &= \sqrt{0 + 25 + 25} \\ &= \sqrt{50}\end{aligned}$$

$$\therefore |\vec{a} + 2\vec{b} - \vec{c}| = 5\sqrt{2}.$$

If l, m and n are the direction cosines of $\vec{a} + 2\vec{b} - \vec{c}$ then

$$l = \frac{0}{5\sqrt{2}}, \quad m = \frac{5}{5\sqrt{2}}, \quad n = \frac{-5}{5\sqrt{2}}$$

$$\therefore l = 0, \quad m = 1/\sqrt{2}, \quad n = -1/\sqrt{2}.$$

5) If $a(1, 0, 0) + b(0, 2, 0) + c(0, 0, 3) = (3, 4, 9)$ then find a, b and c .

$$\rightarrow a(1, 0, 0) + b(0, 2, 0) + c(0, 0, 3) = (3, 4, 9)$$

$$\therefore (a, 0, 0) + (0, 2b, 0) + (0, 0, 3c) = (3, 4, 9)$$

$$\therefore (a+0+0, 0+2b+0, 0+0+3c) = (3, 4, 9)$$

$$\therefore (a, 2b, 3c) = (3, 4, 9)$$

$$\therefore a=3, \quad 2b=4, \quad 3c=9$$

$$\therefore a=3, \quad b=2, \quad c=3$$

6) If $\vec{a} = 3\vec{i} - 2\vec{j} - \sqrt{5}\vec{k}$ and $\vec{b} = 4\vec{i} + 2\vec{j} + \sqrt{5}\vec{k}$ then find the projection of $\vec{a}$ on $\vec{b}$.

$$\rightarrow \text{Here } \vec{a} = (3, -2, -\sqrt{5}), \quad \vec{b} = (4, 2, \sqrt{5})$$

$$\text{Now, } \vec{a} \cdot \vec{b} = (3, -2, -\sqrt{5}) \cdot (4, 2, \sqrt{5})$$

$$= 12 - 4 - 5$$

$$\therefore \vec{a} \cdot \vec{b} = 3.$$

$$\begin{aligned}\rightarrow |\vec{b}| &= \sqrt{(4)^2 + (2)^2 + (\sqrt{5})^2} \\ &= \sqrt{16 + 4 + 5} \\ &= \sqrt{25}\end{aligned}$$

$$\therefore |\vec{b}| = 5$$

$$\rightarrow \text{projection of } \vec{a} \text{ on } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{3}{5}$$

7) If $\vec{a} = \hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$ then find $\vec{a} \cdot (\vec{b} + \vec{c})$.

$$\rightarrow \text{Here } \vec{a} = (1, -1, 1), \vec{b} = (2, -1, 1), \vec{c} = (1, 1, -2)$$

$$\begin{aligned}\text{Now } \vec{b} + \vec{c} &= (2, -1, 1) + (1, 1, -2) \\ &= (3, 0, -1)\end{aligned}$$

$$\begin{aligned}\therefore \vec{a} \cdot (\vec{b} + \vec{c}) &= (1, -1, 1) \cdot (3, 0, -1) \\ &= 3 - 0 - 1\end{aligned}$$

$$\therefore \vec{a} \cdot (\vec{b} + \vec{c}) = 2$$

8) If $\vec{x} = 3\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{y} = 2\hat{i} + \hat{j} - \hat{k}$ then find the vector perpendicular to both $\vec{x}$ and $\vec{y}$.

$$\rightarrow \text{Here } \vec{x} = (3, -1, 2), \vec{y} = (2, 1, -1)$$

$$\begin{aligned}\vec{x} \times \vec{y} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix} \\ &= \hat{i}(1-2) - \hat{j}(-3-4) + \hat{k}(3+2) \\ &= -\hat{i} + 7\hat{j} + 5\hat{k} \\ &= (-1, 7, 5)\end{aligned}$$

$$\begin{aligned}\therefore |\vec{x} \times \vec{y}| &= \sqrt{(-1)^2 + 7^2 + 5^2} \\ &= \sqrt{1 + 49 + 25}\end{aligned}$$

$$\therefore |\vec{x} \times \vec{y}| = \sqrt{75}$$

→ Unit vector perpendicular to both $\vec{x}$ & $\vec{y}$ is

$$\frac{\vec{x} \times \vec{y}}{|\vec{x} \times \vec{y}|} = \frac{(-1, 7, 5)}{\sqrt{75}}$$
$$= \left(-\frac{1}{\sqrt{75}}, \frac{7}{\sqrt{75}}, \frac{5}{\sqrt{75}} \right)$$

9) If $\vec{a} = 10\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} - 2\hat{j} - 2\hat{k}$
then find $\vec{a} \cdot (\vec{b} \times \vec{c})$.

→ Here $\vec{a} = (10, 2, 3)$, $\vec{b} = (1, -2, 2)$, $\vec{c} = (3, -2, 2)$

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix}$$

$$= \hat{i}(4+4) - \hat{j}(-2-6) + \hat{k}(-2+6)$$

$$= 8\hat{i} + 8\hat{j} + 4\hat{k}$$

$$\therefore \vec{a} \cdot (\vec{b} \times \vec{c}) = (10, 2, 3) \cdot (8, 8, 4)$$

$$= 80 + 16 + 12$$

$$\therefore \vec{a} \cdot (\vec{b} \times \vec{c}) = 108$$

10) For what value of m , the vectors $2\hat{i} - 3\hat{j} + 5\hat{k}$ and $m\hat{i} - 6\hat{j} - 8\hat{k}$ are perpendicular to each other?

→ Let $\vec{a} = (2, -3, 5)$, $\vec{b} = (m, -6, -8)$

Here $\vec{a}$ & $\vec{b}$ vectors are perpendicular to each other.

$$\therefore \vec{a} \cdot \vec{b} = 0$$

$$\therefore (2, -3, 5) \cdot (m, -6, -8) = 0$$

$$\therefore (2)(m) + (-3)(-6) + (5)(-8) = 0$$

$$\therefore 2m + 18 - 40 = 0$$

$$\therefore 2m - 22 = 0$$

$$\therefore 2m = 22$$

$$\therefore \boxed{m = 11}$$

11) For $\vec{x} = (-4, 9, 6)$, $\vec{y} = (0, 7, 10)$ and $\vec{z} = (-1, 6, 6)$

Show that $(\vec{x} - \vec{z}) \cdot (\vec{y} - \vec{z}) = 0$.

→ Here $\vec{x} = (-4, 9, 6)$, $\vec{y} = (0, 7, 10)$, & $\vec{z} = (-1, 6, 6)$.

$$\begin{aligned}\vec{x} - \vec{z} &= (-4, 9, 6) - (-1, 6, 6) \\ &= (-4+1, 9-6, 6-6) \\ &= (-3, 3, 0)\end{aligned}$$

$$\begin{aligned}\vec{y} - \vec{z} &= (0, 7, 10) - (-1, 6, 6) \\ &= (0+1, 7-6, 10-6) \\ &= (1, 1, 4)\end{aligned}$$

$$\begin{aligned}\text{Now, } (\vec{x} - \vec{z}) \cdot (\vec{y} - \vec{z}) &= (-3, 3, 0) \cdot (1, 1, 4) \\ &= -3 + 3 + 0\end{aligned}$$

$$(\vec{x} - \vec{z}) \cdot (\vec{y} - \vec{z}) = 0$$

12) Show that the angle between the vectors $2\hat{i} + \hat{j} + 4\hat{k}$ and $\hat{i} + \hat{j} + \hat{k}$ is $\cos^{-1} \frac{\sqrt{7}}{3}$.

→ Let $\vec{a} = 2\hat{i} + \hat{j} + 4\hat{k}$ & $\vec{b} = \hat{i} + \hat{j} + \hat{k}$
 $\vec{a} = (2, 1, 4)$ & $\vec{b} = (1, 1, 1)$

$$\begin{aligned}\therefore |\vec{a}| &= \sqrt{(2)^2 + 1 + 4^2} & |\vec{b}| &= \sqrt{1+1+1} \\ &= \sqrt{4+1+16} & \therefore |\vec{b}| &= \sqrt{3} \\ &= \sqrt{21}\end{aligned}$$

$$\vec{a} \cdot \vec{b} = (2, 1, 4) \cdot (1, 1, 1)$$

$$= 2 + 1 + 4$$

$$\therefore \vec{a} \cdot \vec{b} = 7$$

If θ is the angle between $\vec{a}$ & $\vec{b}$ then

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| \cdot |\vec{b}|}$$

$$= \frac{7}{\sqrt{21} \cdot \sqrt{3}}$$

$$= \frac{\sqrt{7} \cdot \sqrt{7}}{\sqrt{7} \sqrt{3} \sqrt{3}}$$

$$= \frac{\sqrt{7}}{3}$$

$$\therefore \theta = \cos^{-1} \frac{\sqrt{7}}{3}$$

13> show that the angle between the vectors $\hat{i} + \hat{j} + \hat{k}$

and $2\hat{i} - 2\hat{j} + \hat{k}$ is $\sin^{-1} \sqrt{\frac{26}{27}}$

→ Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ & $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$

$$\vec{a} = (1, 1, 1) \quad \vec{b} = (2, -2, 1)$$

Then

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 2 & -2 & 1 \end{vmatrix}$$

$$= \hat{i}(1-2) - \hat{j}(1+2) + \hat{k}(-2-2)$$

$$= -\hat{i} - 3\hat{j} - 4\hat{k}$$

$$\therefore |\vec{a} \times \vec{b}| = \sqrt{(-1)^2 + (-3)^2 + (-4)^2}$$

$$= \sqrt{1+9+16} = \sqrt{26}$$

$$\therefore |\vec{a} \times \vec{b}| = \sqrt{26}$$

$$\therefore |\vec{a}| = \sqrt{1+1+1}$$

$$= \sqrt{3}$$

$$|\vec{b}| = \sqrt{(2)^2 + (-2)^2 + 1^2}$$

$$= \sqrt{4+4+1}$$

$$|\vec{b}| = \sqrt{9}$$

If θ is the angle between two vectors $\vec{a}$ & $\vec{b}$ is

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$$

$$= \frac{\sqrt{26}}{\sqrt{3} \sqrt{9}}$$

$$\therefore \theta = \sin^{-1} \frac{\sqrt{26}}{\sqrt{27}}$$

$$\therefore \theta = \sin^{-1} \sqrt{\frac{26}{27}}$$

14) Find a unit vector perpendicular to each other

$$\vec{a} = (5, 7, -2) \text{ and } \vec{b} = (3, 1, -2)$$

$$\Rightarrow \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 7 & -2 \\ 3 & 1 & -2 \end{vmatrix}$$

$$= \hat{i}(-14+2) - \hat{j}(-10+6) + \hat{k}(5-21)$$

$$= -12\hat{i} + 4\hat{j} - 16\hat{k}$$

$$\therefore |\vec{a} \times \vec{b}| = \sqrt{(-12)^2 + (4)^2 + (-16)^2}$$

$$= \sqrt{144 + 16 + 256}$$

$$|\vec{a} \times \vec{b}| = \sqrt{416} = 4\sqrt{26}$$

- Unit vector perpendicular to both $\vec{a}$ & $\vec{b}$ is

$$\begin{aligned} & \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} \\ &= \frac{(-12, 4, -16)}{4\sqrt{16}} \\ &= \frac{4(-3, 1, -4)}{4\sqrt{16}} \\ &= \frac{(-3, 1, -4)}{\sqrt{16}} \end{aligned}$$

* Question for 4 Marks

1). If $\vec{x} = (1, 1, 1)$ and $\vec{y} = (2, -1, -1)$ then P.T. $\vec{x}$ is perpendicular to $\vec{y}$. Also find a unit vector perpendicular to both $\vec{x}$ and $\vec{y}$.

→ Here $\vec{x} = (1, 1, 1)$ & $\vec{y} = (2, -1, -1)$.

$$\begin{aligned} \text{Now, } \vec{x} \cdot \vec{y} &= (1, 1, 1) \cdot (2, -1, -1) \\ &= 2 - 1 - 1 \end{aligned}$$

$$\therefore \vec{x} \cdot \vec{y} = 0$$

so, $\vec{x}$ is perpendicular to $\vec{y}$.

$$\vec{x} \times \vec{y} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & -1 & -1 \end{vmatrix}$$

$$= \hat{i}(-1+1) - \hat{j}(-1-2) + \hat{k}(-1-2)$$

$$= 0\hat{i} + 3\hat{j} - 3\hat{k}$$

$$= (0, 3, -3)$$

$$|\vec{x} \times \vec{y}| = \sqrt{0+9+9} = \sqrt{18} = 3\sqrt{2}$$

→ Unit vector perpendicular to both $\vec{x}$ & $\vec{y}$ is $\frac{\vec{x} \times \vec{y}}{|\vec{x} \times \vec{y}|}$

$$= \frac{(0, 3, -3)}{3\sqrt{2}}$$

$$= \frac{(0, 1, -1)}{\sqrt{2}}$$

2) If $\vec{a} = 2\vec{i} - 3\vec{j} + 4\vec{k}$ and $\vec{b} = \vec{i} - \vec{j} + \vec{k}$ then find unit vector perpendicular to $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$.

→ Here $\vec{a} = (2, -3, 4)$ and $\vec{b} = (1, -1, 1)$

$$\text{Then } \vec{a} + \vec{b} = (2, -3, 4) + (1, -1, 1) \\ = (3, -4, 5)$$

$$\text{And } \vec{a} - \vec{b} = (2, -3, 4) - (1, -1, 1) \\ = (2-1, -3+1, 4-1) \\ = (1, -2, 3)$$

Unit vector perpendicular to $\vec{a} + \vec{b}$ & $\vec{a} - \vec{b}$ is

$$\frac{(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})}{|(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})|}$$

$$\rightarrow (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & -4 & 5 \\ 1 & -2 & 3 \end{vmatrix}$$

$$= \vec{i}(-12+10) - \vec{j}(9-5) + \vec{k}(-6+4)$$

$$= -2\vec{i} - 4\vec{j} - 2\vec{k}$$

$$= (-2, -4, -2)$$

$$|(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})| = \sqrt{4+16+4} \\ = \sqrt{24}$$

$$= 2\sqrt{6}$$

Unit vector perpendicular to $\vec{a} + \vec{b}$ & $\vec{a} - \vec{b}$ is $\frac{(-2, -4, -2)}{2\sqrt{6}}$

$$= \frac{(-1, -2, -1)}{\sqrt{6}}$$

3> A body is acted upon the forces $3\hat{i}-2\hat{j}+\hat{k}$ and $-\hat{i}-\hat{j}+2\hat{k}$. If the body moves under the forces from the point $(2, 2, -3)$ to $(-1, 2, 4)$ then find work done.

→ Let $\vec{F}_1 = (3, -2, 1)$, & $\vec{F}_2 = (-1, -1, 2)$

$$\begin{aligned}\text{Resultant force } \vec{F}_R &= \vec{F}_1 + \vec{F}_2 \\ &= (3, -2, 1) + (-1, -1, 2) \\ &= (2, -3, 3)\end{aligned}$$

$$d_1 = (2, 2, -3), \quad d_2 = (-1, 2, 4)$$

$$\begin{aligned}\text{displacement } d &= d_2 - d_1 \\ &= (-1, 2, 4) - (2, 2, -3) \\ &= (-3, 0, 7)\end{aligned}$$

$$\begin{aligned}\text{work done } W &= \vec{F}_R \cdot d \\ &= (2, -3, 3) \cdot (-3, 0, 7) \\ &= -6 + 0 + 21\end{aligned}$$

$$\therefore \boxed{W = 15 \text{ Unit}}$$

4> A body is acted upon the forces $3\hat{i}-2\hat{j}+3\hat{k}$ and $-\hat{j}+2\hat{k}$. If the body moves under the forces from the point $(2, 0, -3)$ to $(-1, 2, 2)$ then find work done.

→ Let $\vec{F}_1 = (3, -2, 3)$, $\vec{F}_2 = (0, -1, 2)$

$$d_1 = (2, 0, -3) \quad d_2 = (-1, 2, 2)$$

$$\begin{aligned}\text{Resultant Force } F &= \vec{F}_1 + \vec{F}_2 \\ &= (3, -2, 3) + (0, -1, 2) \\ &= (3, -3, 5)\end{aligned}$$

$$\begin{aligned}\text{displacement } d &= d_2 - d_1 \\ &= (-1, 2, 2) - (2, 0, -3) \\ &= (-3, 2, 5)\end{aligned}$$

$$\begin{aligned} \rightarrow \text{Total work done } W &= \mathbf{F} \cdot \mathbf{d} \\ &= (3, -3, 5) \cdot (-3, 2, 5) \\ &= -9 - 6 + 25 \\ \therefore \boxed{W = 10 \text{ unit}} \end{aligned}$$

5> Forces $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{i} + 3\mathbf{j} - \mathbf{k}$ are acting on a particle and the particle moves from $2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ to the point $5\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ under these forces. Find the work done by the force.

$$\begin{aligned} \rightarrow \text{Here } \mathbf{F}_1 &= (3, -1, 2), \quad \mathbf{F}_2 = (1, 3, -1) \\ \mathbf{d}_1 &= (2, 3, 1), \quad \mathbf{d}_2 = (5, 2, 3) \end{aligned}$$

$$\begin{aligned} \text{Resultant Force } \mathbf{F}_R &= \mathbf{F}_1 + \mathbf{F}_2 \\ &= (3, -1, 2) + (1, 3, -1) \\ &= (4, 2, 1) \end{aligned}$$

$$\begin{aligned} \text{Displacement } \mathbf{d} &= \mathbf{d}_2 - \mathbf{d}_1 \\ &= (5, 2, 3) - (2, 3, 1) \\ &= (3, -1, 2) \end{aligned}$$

$$\begin{aligned} \text{work done } W &= \mathbf{F}_R \cdot \mathbf{d} \\ &= (4, 2, 1) \cdot (3, -1, 2) \\ &= 12 - 2 + 2 \\ &= 12 \end{aligned}$$

$$\therefore \boxed{W = 12 \text{ unit}}$$

6> A particle moves from the point $3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ to the point $\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ under the effect of constant forces $\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $2\mathbf{i} + 5\mathbf{j} - 6\mathbf{k}$. Find the work done.

$$\rightarrow \text{Let } F_1 = (1, -1, 1), F_2 = (1, 1, -3), F_3 = (4, 5, -6)$$

$$\text{Resultant Forces } F_R = \bar{F}_1 + \bar{F}_2 + \bar{F}_3$$

$$= (1, -1, 1) + (1, 1, -3) + (4, 5, -6)$$

$$= (1+1+4, -1+1+5, 1-3-6)$$

$$= (6, 5, -8)$$

$$\text{displacement } d = (1, 3, -4) - (3, -2, 1)$$

$$= (1-3, 3+2, -4-1)$$

$$d = (-2, 5, -5)$$

$$\text{Work done } W = \bar{F}_R \cdot d$$

$$= (6, 5, -8) \cdot (-2, 5, -5)$$

$$= -12 + 25 + 40$$

$$\boxed{W = 53 \text{ Unit}}$$

7) A Force $F = 2i + j + k$ is acting at the point $(-3, 2, 1)$. Find the Magnitude of the moments of Force F about the point $(2, 1, 2)$.

$$\rightarrow \text{Here } \bar{F} = (2, 1, 1).$$

Let $A(2, 1, 2)$ and $P(-3, 2, 1)$ be the given points.

$$\therefore \bar{AP} = \bar{OP} - \bar{OA}$$

$$= (-3, 2, 1) - (2, 1, 2)$$

$$= (-3-2, 2-1, 1-2)$$

$$\bar{AP} = (-5, 1, -1)$$

$$\text{Moment of the force } F \text{ about the point } A = \bar{AP} \times \bar{F}$$

$$= (-5, 1, -1) \times (2, 1, 1)$$

$$= \begin{vmatrix} i & j & k \\ -5 & 1 & -1 \\ 2 & 1 & 1 \end{vmatrix}$$

$$= i(1+1) - j(-5+2) + k(-5-2)$$

$$= 2i + 3j - 7k$$

$$\vec{AP} \times \vec{F} = (2, 3, -7)$$

$$\begin{aligned} \text{Magnitude of the moment} &= \sqrt{(2)^2 + (3)^2 + (-7)^2} \\ &= \sqrt{4 + 9 + 49} \\ &= \sqrt{62}. \end{aligned}$$

8). Find the moment about the point $(2, 3, -1)$ of the force $3i - k$ acting through the point $(1, -2, 1)$. Also find the magnitude of the moment.

$$\rightarrow \text{Let } \vec{F} = 3i - k, \quad \vec{F} = (3, 0, -1)$$

Let $A(2, 3, -1)$ and $P(1, -2, 1)$ be the given points.

$$\therefore \vec{AP} = \vec{OP} - \vec{OA}$$

$$= (1, -2, 1) - (2, 3, -1)$$

$$= (1-2, -2-3, 1+1)$$

$$= (-1, -5, 2).$$

Moment of the Force F about the point $A = \vec{AP} \times \vec{F}$

$$= (-1, -5, 2) \times (3, 0, -1)$$

$$= \begin{vmatrix} i & j & k \\ -1 & -5 & 2 \\ 3 & 0 & -1 \end{vmatrix}$$

$$\begin{aligned} &= i(5-0) - j(1-6) + k(0+15) = 5i + 5j + 15k \\ &= (5, 5, 15) \end{aligned}$$

$\therefore$ Magnitude of the moment $= |\vec{AP} \times \vec{F}|$

$$= \sqrt{(5)^2 + (5)^2 + (15)^2}$$

$$= \sqrt{25 + 25 + 225} = \sqrt{275} = 5\sqrt{11}$$